

Affine-invariant projection methods for conservative integration of differential equations

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Abstract

Numerical integrators which conserve invariants of original differential equations are called conservative integrators. While most of them are computationally expensive, a cheaper class is the projection methods, where conservation is realized by simple projections. However, such methods do not necessarily preserve affine invariance originally held in differential equations. In this study, we point out several affine-invariant projections and discuss how the affine invariance affects the quality of numerical solutions.

Keywords ordinary differential equations, numerical methods, conservation, affine-invariance

Research Activity Group Scientific Computation and Numerical Analysis

1. Introduction

In this study, we are concerned with the numerical integration of the following autonomous system on $x: \mathbb{R}^+ \rightarrow \mathbb{R}^d$:

$$\begin{cases} \dot{x} = f(x), \\ x(0): \text{given.} \end{cases} \quad (1)$$

Suppose, furthermore, that the system has an invariant $I(x(t)) = I_0$. Then it is more desirable to preserve it in discrete setting. Such integrators are called conservative integrators, and have been proved to be effective in various application areas including astronomy, molecular dynamics and physics [1]. The most successful class of such integrators is the discrete gradient (DG) method [2], which provides generic conservative integrators for skew-gradient systems. However it has a severe drawback that resulting schemes are generally fully-implicit and thus computationally expensive. Another general way for designing conservative integrators is the projection methods, where conservation is forced by simply projecting numerical solutions onto the manifold defined by the target invariant [1]. This is more primitive (compared to DG) and can occasionally be unstable. Conversely, it has virtues that it can be applied to any systems with invariants, and more importantly, that it only requires solving scalar nonlinear equation at every step, which is much cheaper than DG.

Another important feature in numerical methods is the affine invariance, which requires numerical methods to be compatible to affine coordinate transformations. This is natural especially when the problem allows arbitrary choice of the coordinate system within affine transformations. Many popular general-purpose methods are in fact affine invariant [1, IV.1]. However, projection methods are not necessarily so; for example, the most

popular orthogonal projection (explained later) is not affine invariant.

The aim of this study is to discuss the possibility of affine-invariant projection methods. We first show that projection methods can be affine invariant. In particular, we propose a new projection method utilizing the Newton direction. Then we show some numerical examples which illustrate that affine invariance can in fact improve the quality of projection methods.

2. Existing projection methods

Projection methods are formalized as follows. Let $\psi: \mathbb{R}^d \rightarrow \mathbb{R}^d$ be the base method (arbitrary numerical scheme for (1)), and $D_n \in \mathbb{R}^d \setminus \{0\}$ be the *direction of projection*. We denote the numerical solutions by $x_n (n = 0, 1, \dots)$, $x_0 := x(0)$. Then a projection method can be written as

$$\hat{x}_{n+1} = \psi(x_n),$$

$$x_{n+1} = \hat{x}_{n+1} + \lambda D_n, \lambda \in \mathbb{R}, I(x_{n+1}) = I_0.$$

When the base method ψ is explicit, the main effort lies in solving the scalar nonlinear equation on λ .

Below we are mainly concerned with the direction D_n 's. The following are some existing examples.

2.1 Orthogonal projection

In this case we take the following direction. (This is actually an approximation of mathematical “orthogonal projection” in the strict sense of the word. See [1] for the detail. Similarly, other “projections” below are simply suggesting directions to reach the manifold.)

$$D_n = \nabla I(\hat{x}_{n+1}). \quad (2)$$

2.2 Projections with two solutions

Another simple way is to first construct two tentative solutions by two methods, ψ and ψ' , and determine the

direction by the following (Calvo et al. [3]).

$$D_n = \psi'(x_n) - \hat{x}_{n+1}. \quad (3)$$

For reducing computational complexity, ψ and ψ' are chosen to form an embedded Runge–Kutta method.

3. Affine invariance

Let us consider the affine transformation $y = Ax + b$, where $A \in \mathbb{R}^{d \times d}$ is an invertible matrix and $b \in \mathbb{R}^d$. The equation (1) is then transformed as

$$\begin{cases} \dot{y} = Af(A^{-1}(y - b)) =: \tilde{f}(y), \\ y(0) = Ax_0 + b. \end{cases} \quad (4)$$

The system (4) has the transformed invariant $\tilde{I}(y) := I(A^{-1}(y - b))$, since

$$\begin{aligned} \frac{d}{dt} \tilde{I}(y) &= \nabla \tilde{I}(y(t))^\top \tilde{f}(y) \\ &= \nabla I(A^{-1}(y - b))^\top A^{-1} A f(A^{-1}(y - b)) = 0. \end{aligned}$$

The last equality is from $\dot{I} = (\nabla I(x))^\top f(x) = 0$.

Next, let us define the affine invariance of conservative numerical method for (1). To this end, let us consider a general method:

$$\begin{cases} x_{n+1} = \Phi(f, I)(x_n) \quad (n = 0, 1, \dots), \\ x_0 = x(0). \end{cases}$$

The notation $\Phi(f, I)$ means the conservative integrator depend also on the target invariant I ; this notation also clarifies the effect of transformation. Applying the same numerical method to the transformed equation (4) to find y_n approximating $y(n\Delta t)$, we obtain

$$\begin{cases} y_{n+1} = \Phi(\tilde{f}, \tilde{I})(y_n), \\ y_0 = Ax_0 + b. \end{cases} \quad (5)$$

Definition 1 A numerical scheme Φ is said to be affine-invariant if the following condition holds:

$$\Phi(\tilde{f}, \tilde{I})(Ax + b) = A\Phi(f, I)(x) + b. \quad (6)$$

If (6) holds, then the two solutions are consistent in the sense that $y_n = Ax_n + b$ ($n = 0, 1, \dots$) holds. This is the affine invariance we aim at. For example, all Runge–Kutta methods are affine-invariant [1].

Now suppose the scheme Φ is a projection method consisting of a base method Ψ and a projection step with the direction D_n 's:

$$\begin{cases} \Phi(f, I)(x_n) = \hat{x}_{n+1} + \lambda D(x_n, \hat{x}_{n+1}, I), \\ \text{where } \hat{x}_{n+1} = \Psi(f)(x_n), \\ \lambda \in \mathbb{R} \text{ s.t. } I(\hat{x}_{n+1} + \lambda D(x_n, \hat{x}_{n+1}, I)) = I_0. \end{cases} \quad (7)$$

Here, to clarify the effect of transformation, we rewrote $\Psi = \Psi(f)$ (we assume the base method is not conservative and does not refer to I) and $D_n = D(x_n, \hat{x}_{n+1}, I)$.

Then the next definition is natural so that the overall projection scheme becomes affine invariant.

Definition 2 A direction of projection D is said to be affine-invariant if the following condition holds:

$$D(Ax_n + b, A\hat{x}_{n+1} + b, \tilde{I}) = AD(x_n, \hat{x}_{n+1}, I).$$

Theorem 3 Suppose the base method Ψ and the direction of projection D are affine-invariant. Then Φ defined in (7) is affine-invariant.

Proof The transformed scheme (5) with the projection scheme (7) reads

$$\begin{cases} \Phi(\tilde{f}, \tilde{I})(y_n) = \hat{y}_{n+1} + \lambda' D(y_n, \hat{y}_{n+1}, \tilde{I}), \\ \text{where } \hat{y}_{n+1} = \Psi(\tilde{f})(y_n), \\ \lambda' \in \mathbb{R} \text{ s.t. } \tilde{I}(\hat{y}_{n+1} + \lambda' D(y_n, \hat{y}_{n+1}, \tilde{I})) = I_0. \end{cases} \quad (8)$$

We start with the pair (x_n, y_n) related by the affine transformation, i.e., $y_n = Ax_n + b$. Then, because we assume Ψ is affine invariant, we have $\hat{y}_{n+1} = A\hat{x}_{n+1} + b$. This, together with the affine invariance of D , implies $D(y_n, \hat{y}_{n+1}, \tilde{I}) = AD(x_n, \hat{x}_{n+1}, I)$. Thus the first equation of (8) can be rewritten as

$$\Phi(\tilde{f}, \tilde{I})(Ax_n + b) = A(\hat{x}_{n+1} + \lambda' D(x_n, \hat{x}_{n+1}, I)) + b.$$

Furthermore, by the definition of $\tilde{I}$, we have

$$\tilde{I}(\hat{y}_{n+1} + \lambda' D(y_n, \hat{y}_{n+1}, \tilde{I})) = I(\hat{x}_{n+1} + \lambda' D(x_n, \hat{x}_{n+1}, I)),$$

which means $\lambda' = \lambda$. This concludes (6). (QED)

In summary, the combination of an affine invariant base method and an affine-invariant projection direction yields an affine-invariant projection method. For the latter, the two-solutions projections (3) are affine-invariant [3], while the orthogonal projection (2) is not. Next we propose another natural choice.

4. Newton direction projection methods

We introduce the following direction. Although we believe this is new (in the context of projection methods), it is analogous to the fact that the Newton method in optimization is affine invariant (see, for example, [4]).

Definition 4 The Newton direction is the direction defined as follows:

$$D_n = D(x_n, \hat{x}_{n+1}, I) = (\nabla^2 I(\hat{x}_{n+1}))^{-1} \nabla I(\hat{x}_{n+1}).$$

Theorem 5 The Newton direction is affine-invariant.

Proof

$$\begin{aligned} D(Ax_n + b, A\hat{x}_{n+1} + b, \tilde{I}) &= (\nabla^2 \tilde{I}(A\hat{x}_{n+1} + b))^{-1} \nabla \tilde{I}(A\hat{x}_{n+1} + b) \\ &= A(\nabla^2 \tilde{I}(\hat{x}_{n+1}))^{-1} A^\top A^{-\top} \nabla I(\hat{x}_{n+1}) \\ &= AD(x_n, \hat{x}_{n+1}, I). \end{aligned} \quad (QED)$$

5. Numerical experiments

We show some numerical examples that illustrate how and to which extent the affine invariance helps the qualitative behavior of projection methods.

5.1 Simple harmonic motion

The following two equations are related by an affine transformation $y_1 = x_1, y_2 = 100x_2$.

$$\dot{x}_1 = -x_2, \quad \dot{x}_2 = x_1. \quad (9)$$

$$\dot{y}_1 = -y_2/100, \quad \dot{y}_2 = 100y_1. \quad (10)$$

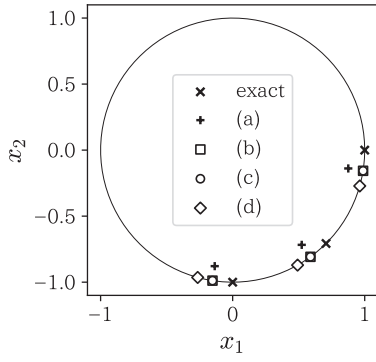


Fig. 1. Solutions of (9).

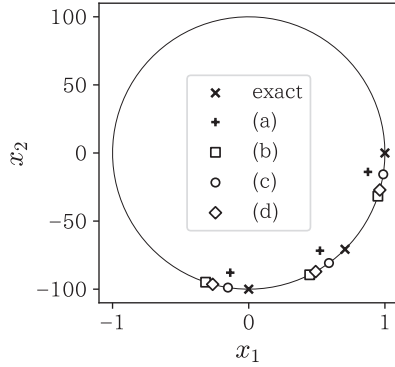


Fig. 2. Solutions of (10).

The system (10) is a typical case where affine invariance is helpful, since the orbit of (10) has points where the curvature becomes far larger than that of (9).

We used the following four methods, which are all based on the Runge–Kutta 3/8 rule:

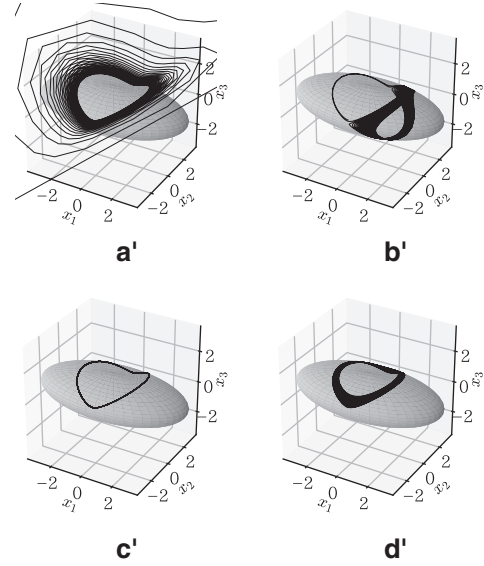
- (a) the Runge–Kutta 3/8 rule (not conservative),
- (b) the rule + orthogonal projection,
- (c) the rule + Newton direction projection,
- (d) projection with two solutions based on an embedded method obtained by adding $(1/12, 1/2, 1/4, 0, 1/6)$ to the Butcher tableau of the Runge–Kutta 3/8 rule (see [1, Section 2.1, Table 1.2]).

The invariants to be conserved by the projection are $x_1^2/2 + x_2^2/2$ and $y_1^2/2 + y_2^2/20,000$.

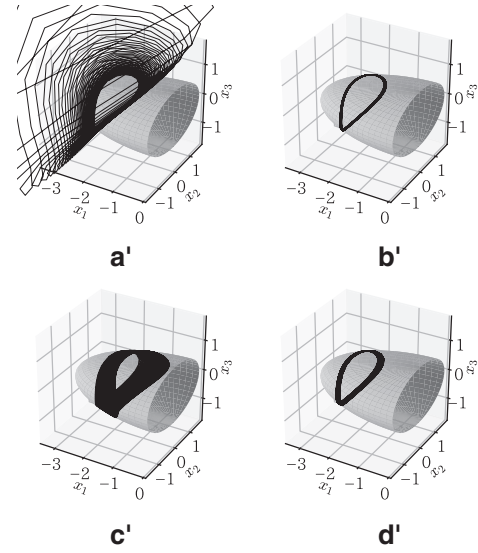
We used the step size $\Delta t = \pi/4$. The initial solutions are $(x_1, x_2) = (1, 0)$ and $(y_1, y_2) = (1, 0)$. The solutions at $t = 78\pi/4, 79\pi/4, 80\pi/4$ are shown in Figs. 1 and 2. In Fig. 1, we see the base method does not preserve the invariant, while others are conservative as expected. The solutions of orthogonal and Newton direction projections are the same. The projection with two solutions slows down the solution heavier. For the anisotropic problem (10), we see the effect of the affine invariance. The solutions of the Newton direction projection stay the same (under the affine transformation), while the solutions of the orthogonal projection are greatly delayed.

5.2 Rigid body

The following equation (11) describe the precession of a rigid body without any torque applied, where a_1, a_2 , and a_3 represent its principal moments of inertia and x_1 ,



Initial point: $(0, -2.3 \cos(0.1), 2.3 \sin(0.1))$



Initial point: $(-2.3 \cos(0.3), 0, 2.3 \sin(0.3))$

Fig. 3. Solutions of (11).

x_2 , and x_3 represent its angular momentum, each based on the body-fixed frame.

$$\dot{x}_i = \frac{a_k - a_j}{a_j a_k} x_j x_k \quad (11)$$

$$((i, j, k) = (1, 2, 3), (2, 3, 1), (3, 1, 2)).$$

The principal moments are here set to $a_1 = 4, a_2 = 1$, and $a_3 = 1/3$ so that the system is anisotropic and affine invariance is hoped.

Fig. 3 shows the numerical solutions of the equation (11). We used the following four methods, which are based on the Heun method:

- (a') the Heun method (not conservative),
- (b') the method + orthogonal projection,
- (c') the method + Newton direction projection,
- (d') projection with two solution based on an embedded method with the Heun method and the explicit Euler method.

In all experiments, the target invariant of the projection is $x_1^2/a_1 + x_2^2/a_2 + x_3^2/a_3$, the Hamiltonian. We used the step size $\Delta t = 0.1$ and calculated between $t = 0$ and $t = 100$. We carried out calculations with two initial solutions: $(x_1, x_2, x_3) = (0, -2.3 \cos(0.1), 2.3 \sin(0.1))$ and $(x_1, x_2, x_3) = (-2.3 \cos(0.3), 0, 2.3 \sin(0.3))$. Their exact solutions are closed orbits circling around the fixed points on the x_3 - and x_1 -axes, respectively. In both cases, the pure Heun method inhibits instability, due to the lack of the conservation. Other methods are conservative as expected, but results in different dynamics.

In the former setting, the Newton direction projection (c') gives the best result; the solution correctly move around the fixed point. The two-solutions projection (d') gives slightly worse result, where the orbit slightly shrinks. The solution of the orthogonal projection (b') is more unphysical, where it transits to another orbit around another fixed point.

In the latter setting, all projection methods resulted in similar solutions, but the orthogonal projection (b') gives the sharpest orbit. The orbit of the Newton direction projection (c') is drifting and worse. This means that the Newton direction projection is not always the best.

5.3 KdV equation

Lastly, we show an example on an ordinary differential equation arising from a partial differential equation (PDE). Let us consider the Korteweg–de Vries equation:

$$u_t = - \left(\frac{u^2}{2} + \varepsilon u_{xx} \right)_x \quad (12)$$

on $x \in [0, L]$ under the periodic boundary condition. (Here we abuse the symbol x , which is now the space variable. We employ this notation because this is standard in PDE setting, and no confusion occurs.) Its Hamiltonian is $\int_0^L (\frac{1}{6}u^3 - \frac{\varepsilon}{2}u_x^2) dx$. One semi-discretized version is

$$\dot{U} = - \frac{\delta^{(1)}(U^2)}{2} - \varepsilon \delta^{(3)}U, \quad (13)$$

where $\delta^{(1)}$, $\delta^{(3)}$ are the standard first and third order central finite differences and δ^+ is the forward finite difference. If we divide the spatial interval $[0, L]$ into N grids, the equation (13) has the associated discrete Hamiltonian $H(U) = \sum_{i=1}^N (\frac{1}{6}U_i^3 - \frac{\varepsilon}{2}((\delta^+ U)_i)^2)$, which is to be preserved.

As the initial data, we consider one-soliton solution $u(x, 0) = 3 \operatorname{sech}^2((x - t - L/2)/(2\sqrt{\varepsilon}))$ on $\mathbb{R}$.

We take the coefficient $\varepsilon = 0.022^2$ considered in [5], which is quite small so that the solution vanishes at $x = 0, L$ and we can regard the solution to be almost periodic. We set $L = 2$, and the spatial mesh size to be $L/109$. We set the step size $\Delta t = 0.01$ and integrate numerically between $t = 0$ and $t = 200$. We apply the same methods (a), (b), (c), and (d) as in Section 5.1 to (13).

Figs. 4 and 5 show the numerical solutions.

Let us first consider Fig. 5. There, the error in the positions of the soliton are shown, ideally which should be 0. The positions are numerically computed as weighted average of a numerical solution (in space direction). As

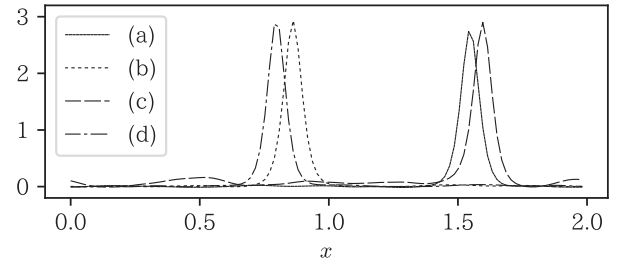


Fig. 4. Solution of (12) at $t = 200$.

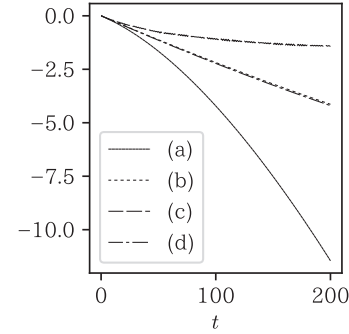


Fig. 5. Error in the positions of the soliton.

a result, the Newton direction (c) achieved the smallest error, and the nonconservative method (a) the biggest. The orthogonal projection (b) and the two solution projection (d) came between the two, showing similar performance.

Fig. 4 shows the snapshot of the solutions at $t = 200$. The exact solution should come back to the original position $x = 1$ (since the speed is 1, and $L = 2$), but as described above the speeds of the numerical solitons are inaccurate, and the aim of this figure is just to check the forms of the solutions. All of the four methods roughly captures the soliton shape even after the long integration.

6. Conclusion

We formally defined the affine invariance of projection methods, and proposed the Newton direction as a new affine-invariant direction of projection. Several numerical experiments ODEs and a PDE showed that affine invariance can improve the quality of projection methods, and the Newton direction projection is also useful in some cases. However, as Fig. 3 shows, the Newton direction is not always the best. Whether another affine invariant direction that always outperform existing directions exists or not is an interesting open question.

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